ADS Experiment 5

Experiment: Performance Evaluation of Clustering Models

AIM

To implement and explore performance evaluation metrics for unsupervised data models (clustering). Specifically, apply K-Means clustering for Customer Segmentation on the Online Retail dataset using RFM features, and evaluate the clustering quality using both:

- Extrinsic measures – Rand Index and Mutual Information
- Intrinsic measure – Silhouette Coefficient

Theory

1. What is Clustering?

Clustering is an unsupervised machine learning technique used to group data points into clusters such that:

- Data points within the same cluster are more similar to each other.
- Data points in different clusters are dissimilar.

Unlike supervised learning, clustering does not require predefined labels. The algorithm automatically discovers patterns or structures in the dataset.

Clustering is widely used in:

- Customer segmentation
- Image segmentation
- Market basket analysis
- Document grouping
- Recommendation systems

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2. Why Evaluate Clustering?

Since clustering is unsupervised, there are no predefined labels to calculate accuracy directly.

Therefore, specialized evaluation metrics are used. These are broadly divided into two categories:

1. Extrinsic Measures

These require ground truth labels and compare predicted clusters with actual labels.

2. Intrinsic Measures

These evaluate clustering based only on the data itself, without any external labels.

Extrinsic Measures

1. Rand Index (RI) and Adjusted Rand Index (ARI)

The Rand Index (RI) measures the similarity between two clusterings by considering all pairs of samples and checking whether they are assigned to the same or different clusters in both predicted and true labels.

Expected Rand Index

$$E[RI] = \frac{\sum_i \binom{a_i}{2} \times \sum_j \binom{b_j}{2}}{\binom{n}{2}}$$

Where:

- a_i= number of samples in true cluster i
- b_j= number of samples in predicted cluster j
- n= total number of samples

Adjusted Rand Index (ARI)

ARI corrects the Rand Index for random chance.

$$ARI = \frac{\sum_{ij} \binom{n_{ij}}{2} - E[RI]}{\frac{1}{2}\left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2}\right] - E[RI]}$$

Where:

- n_{ij}= number of samples common in true cluster i and predicted cluster j

Range of ARI

Value	Meaning
1	Perfect agreement
0	Random clustering

Value	Meaning
< 0	Worse than random

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2. Mutual Information (MI) and Normalized Mutual Information (NMI)

Mutual Information (MI) measures the amount of shared information between two clusterings.

It quantifies how much knowing one clustering reduces uncertainty about the other.

Formula

$$MI(U,V) = \sum_{i=1}^{|U|} \sum_{j=1}^{|V|} \frac{|U_i \cap V_j|}{N} \log \frac{N\,|U_i \cap V_j|}{|U_i|\,|V_j|}$$

Where:

- U and V = two clusterings (true and predicted)
- $|U_i|$ = number of samples in cluster U_i
- $|V_j|$ = number of samples in cluster V_j
- $|U_i \cap V_j|$ = samples present in both clusters
- N = total samples

Normalized Mutual Information (NMI)

To make MI comparable across datasets, it is normalized.

$$NMI = \frac{MI(U,V)}{\sqrt{H(U)H(V)}}$$

Range

Value	Meaning
1	Perfect correlation
0	No shared information

Intrinsic Measure

Silhouette Coefficient

The Silhouette Score measures how similar a data point is to its own cluster compared to other clusters.

Formula

$$s(i) = \frac{b(i) - a(i)}{\max\,(a(i), b(i))}$$

Where:

- a(i) = average distance from sample *i* to all other points in the same cluster (cohesion)
- b(i) = average distance from sample *i* to points in the nearest neighboring cluster (separation)

Interpretation

Value	Meaning
+1	Perfect clustering
0	Points near cluster boundary
-1	Points assigned to wrong cluster

The overall Silhouette Score is the average of all sample scores.

RFM Analysis for Customer Segmentation

RFM is a popular technique used in customer behavior analysis.

RFM stands for:

Recency (R)
How recently a customer made a purchase.
Lower value indicates more recent activity.

Frequency (F)
How often a customer makes purchases.
Higher value indicates loyal customers.

Monetary (M)
Total amount spent by the customer.
Higher value indicates high-value customers.

These three features create a compact behavioral profile for each customer, making them ideal for clustering.

Conclusion

For the Online Retail Dataset, clustering was applied to perform customer segmentation using RFM features.

Customer Segmentation

The dataset contains transactional data such as:

- InvoiceNo
- Quantity
- UnitPrice
- CustomerID
- InvoiceDate

By computing RFM features per customer, raw transaction data was transformed into meaningful behavioral profiles.

Using K-Means clustering, three major customer segments were identified:

1. High-value customers
2. Mid-value customers
3. Low-value customers

Evaluation of Clustering

The clustering performance was evaluated using both intrinsic and extrinsic metrics.

Silhouette Coefficient

- Measured how well-separated the clusters are.
- Confirmed that customer groups were distinct in RFM feature space.

Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI)

- Used a Monetary-based high/low spender split as proxy ground truth.
- Verified that K-Means clusters align with actual spending behavior.

Business Impact

Customer segmentation provides valuable insights for businesses:

- Retain high-value customers through loyalty programs
- Re-engage inactive customers with targeted promotions
- Upsell mid-value customers to increase lifetime value
- Optimize marketing budget allocation

Importance of Multiple Metrics

Using only intrinsic metrics (like Silhouette Score) would evaluate cluster geometry only.

Adding extrinsic metrics (ARI and NMI) confirms whether clusters correspond to real-world business behavior, making the evaluation more reliable and meaningful.